

Upload the dataset files

In [2]: `from google.colab import files`

`uploaded = files.upload()`

No file chosen

Upload widget is only available when the cell has been executed in the current browser session.

Please rerun this cell to enable.

Saving codetable.txt to codetable.txt

Saving SouthGermanCredit.asc to SouthGermanCredit (1).asc

Load and preview the dataset

In [5]: `import pandas as pd`

`df = pd.read_csv("SouthGermanCredit.asc", sep=r"\s+")`

`df.head()`

Out[5]:

	laufkont	laufzeit	moral	verw	hoehe	sparkont	beszeit	rate	famges	buerge	...	verm	alter	weatkred	wohn	bishkred	b
0	1	18	4	2	1049	1	2	4	2	1	...	2	21	3	1	1	
1	1	9	4	0	2799	1	3	2	3	1	...	1	36	3	1	2	
2	2	12	2	9	841	2	4	2	2	1	...	1	23	3	1	1	
3	1	12	4	0	2122	1	3	3	3	1	...	1	39	3	1	2	
4	1	12	4	0	2171	1	3	4	3	1	...	2	38	1	2	2	

5 rows × 21 columns



Display dataset information

In [8]: `df.info()`

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 21 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   laufkont    1000 non-null   int64
 1   laufzeit    1000 non-null   int64
 2   moral       1000 non-null   int64
 3   verw        1000 non-null   int64
 4   hoehe       1000 non-null   int64
 5   sparkont    1000 non-null   int64
 6   beszeit     1000 non-null   int64
 7   rate        1000 non-null   int64
 8   famges      1000 non-null   int64
 9   buerge      1000 non-null   int64
10   wohnzeit    1000 non-null   int64
11   verm        1000 non-null   int64
12   alter       1000 non-null   int64
13   weitekred   1000 non-null   int64
14   wohn        1000 non-null   int64
15   bishkred    1000 non-null   int64
16   beruf       1000 non-null   int64
17   pers        1000 non-null   int64
18   telef       1000 non-null   int64
19   gastarb     1000 non-null   int64
20   kredit      1000 non-null   int64
dtypes: int64(21)
memory usage: 164.2 KB

```

Display the names of all variables

```
In [14]: df.columns
```

```
Out[14]: Index(['laufkont', 'laufzeit', 'moral', 'verw', 'hoehe', 'sparkont', 'beszeit',
               'rate', 'famges', 'buerge', 'wohnzeit', 'verm', 'alter', 'weitekred',
               'wohn', 'bishkred', 'beruf', 'pers', 'telef', 'gastarb', 'kredit'],
              dtype='object')
```

Separate the predictor variables and the target variable

```
In [15]: X = df.drop("kredit", axis=1)

y = df["kredit"]
```

Split the dataset into training and testing sets

```
In [16]: from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42)
```

Visualize the trained Decision Tree model

```
In [19]: import matplotlib.pyplot as plt
from sklearn.tree import plot_tree

plt.figure(figsize=(20, 10))

plot_tree(
    model,
    feature_names=X.columns,
    class_names=["1", "2"],
    filled=True,
    rounded=True,
    fontsize=8)

plt.show()
```

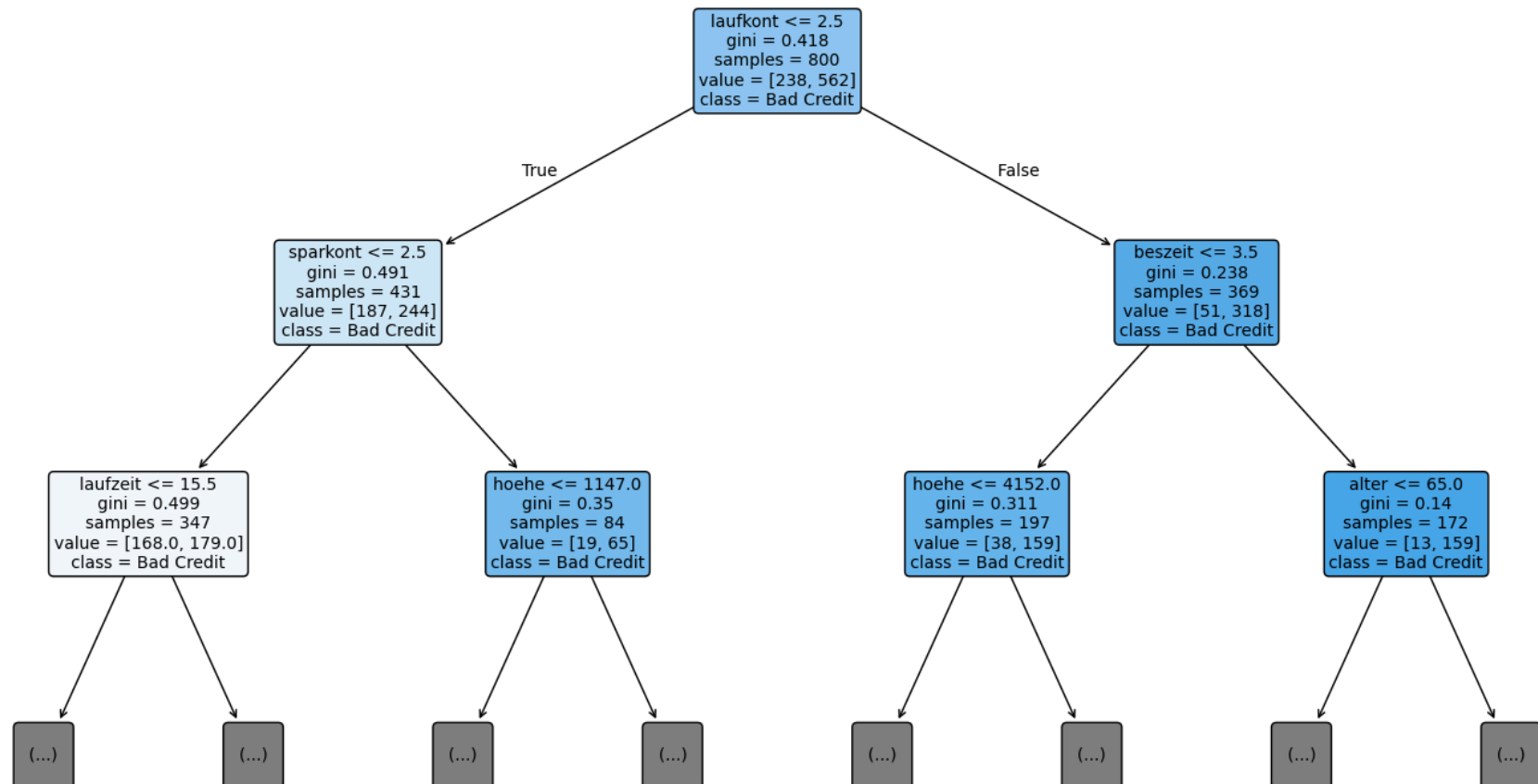


file:///C:/Users/ele/Downloads/Untitled0.html

```

fontsize=10)
plt.show()

```



First Split The root node splits the data using `laufkont`, which represents the status of the checking account. Customers with a value less than or equal to 2.5 follow the left branch, while customers with a value greater than 2.5 follow the right branch. This indicates that checking account status is the most important variable for separating the two credit classes.

Second Split At the second level, the left branch is split using sparkont (savings account status), while the right branch is split using beszeit (employment duration). These variables provide additional information for classifying customers after the initial split based on checking account status.

Calculate and sort feature importance

```
In [30]: importance = pd.DataFrame({
    "Feature": X.columns,
    "Importance": model.feature_importances_})
importance = importance.sort_values(
    by="Importance",
    ascending=False)
importance
```

Out[30]:

	Feature	Importance
4	hoehe	0.188050
1	laufzeit	0.127252
12	alter	0.125079
0	laufkont	0.111879
3	verw	0.078326
2	moral	0.054677
6	beszeit	0.038546
10	wohnzeit	0.034252
9	buerge	0.032754
7	rate	0.031378
5	sparkont	0.029907
15	bishkred	0.028395
13	weitkred	0.027039
8	famges	0.025067
17	pers	0.016851
11	verm	0.015223
16	beruf	0.012265
19	gastarb	0.010253
14	wohn	0.008322
18	telef	0.004486

Most Significant Variables:

The most important variable is hoehe, with a feature importance of 0.188. The next most important variables are laufzeit, alter, laufkont, and verw. These variables contribute the most to the decision tree when classifying customers into different credit risk categories.